LSLFormer: A Lightweight Spectral–LiDAR Fusion Network for Remote Sensing Image Classification

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***Abstract*—The fusion of hyperspectral images (HSIs) and light detection and ranging (LiDAR) data provides rich complementary spectral and elevation information for land cover classification. However, existing fusion methods, particularly Transformer-based models, are often constrained by high computational costs and complex cross-modal interaction mechanisms. To address this challenge, we propose a lightweight spectral–LiDAR fusion network (LSLFormer), which aims to achieve efficient and accurate remote sensing image classification. We introduce three key modules into LSLFormer architecture: 1) a hyperspectral-to-multispectral (H2M) module to alleviate the computational burden of self-attention on high-dimensional spectral data; 2) a multiscale channel interaction and enhancement (MCIE) module to extract robust spatial–structural features from LiDAR data; and 3) a spectral–LiDAR attention (SLA) module to achieve deep cross-modal interaction by dynamically fusing normalized spectral and structural affinity distributions. In addition, we also propose a novel cross-modal structural consistency (CMSC) loss. This mechanism aligns the geometric topology of the spectral features with the LiDAR structure via knowledge distillation, ensuring precise boundary delineation without compromising spectral semantics. Extensive experiments on three public benchmark datasets have demonstrated that LSLFormer consistently outperforms other state-of-the-art (SOTA) convolutional and Transformer-based methods in terms of classification accuracy and computational cost. The codes of this work will be available at https://github.com/DianLi2002/LSLFormer**

***Index Terms*—Data fusion, hyperspectral images (HSIs), light detection and ranging (LiDAR), lightweight, remote sensing, Transformer.**

I. Introduction

REMOTE sensing image classification is a fundamental task in Earth observation, providing crucial technical support for various applications such as urban planning, environmental monitoring, resource management, and precision agriculture [1], [2]. Among the various data sources, hyperspectral images (HSIs) and light detection and ranging (LiDAR) data have drawn significant attention. HSI captures hundreds of narrow, continuous spectral bands, providing rich information for distinguishing different land cover materials [3], [4]. However, HSIs often have limited spatial resolution, and their numerous bands result in high computational requirements and redundant data [5]. Conversely, LiDAR data can provide precise elevation and structural information about the study area at any time of the day and under adverse weather conditions. This offers excellent complementarity to the rich spectral information of HSI [6]. Consequently, the fusion of HSI and LiDAR data has become a prominent and effective research direction [7], [8]. This approach leverages their complementary characteristics to achieve more accurate and robust classification results than using either modality alone.

In recent years, deep learning has become the dominant methodology for HSI and LiDAR fusion. Convolutional neural networks (CNNs), in particular, have demonstrated remarkable success. Models based on 2D-CNNs [9], 3D-CNNs [10], or hybrid architectures (e.g., HybridSN [11]) have shown powerful capabilities in extracting local spatial–spectral features. However, CNNs are overly concerned with spatial content information and can hardly be capable of capturing the spectral sequence attributes well.

More recently, the Transformer architecture [12], originally developed for natural language processing, has demonstrated outstanding performance in computer vision tasks. Vision Transformers (ViTs) [13] and their variants leverage the self-attention mechanism to effectively capture global dependencies, leading to state-of-the-art (SOTA) results in remote sensing classification [14]. Despite their success, Transformers introduce a significant challenge: the self-attention mechanism has a computational complexity that is quadratic with respect to the input sequence length [15], [16]. When applied to high-dimensional HSI data (with hundreds of spectral bands) or large spatial patches, this results in prohibitive computational costs and high memory consumption. Furthermore, designing effective and efficient cross-modal fusion mechanisms for HSI and LiDAR within a Transformer framework remains a complex and open research problem.

To address these limitations, this article proposes a lightweight spectral–LiDAR fusion network (LSLFormer). The primary objective of LSLFormer is to achieve an optimal balance between classification accuracy and computational efficiency. First, drawing inspiration from the optical integration mechanism of multispectral sensors, we propose a

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hyperspectral-to-multispectral (H2M) module. Unlike standard dimensionality reduction (DR) techniques (e.g., PCA [17]), H2M learns a set of "virtual" spectral response functions (SRFs), allowing the network to adaptively aggregate redundant bands into discriminative features [18], [19]. Second, to effectively extract salient spatial–structural features from the LiDAR data prior to fusion, we design a multiscale channel interaction and enhancement (MCIE) module. Third, to resolve the modality mismatch in attention mechanisms, we design a spectral–LiDAR attention (SLA) module. It normalizes and fuses the spectral correlation matrix with the LiDAR-derived structural affinity matrix, enabling the network to dynamically weigh spectral similarity against geometric continuity. Finally, we devise a novel cross-modal structural consistency (CMSC) loss function. This loss functions as a structural regularizer, enforcing the HSI features to mimic the pixel-to-pixel topological relationships found in the LiDAR data.

The main contributions are summarized as follows.

1) We propose LSLFormer, a novel and lightweight Transformer-based network for HSI and LiDAR data fusion that achieves an SOTA balance between classification accuracy and computational cost.
2) We design the H2M module for physically inspired spectral aggregation, the MCIE module for robust multiscale LiDAR feature extraction, and the SLA module for normalized and dual-stream cross-modal attention fusion.
3) We introduce the CMSC loss, a novel structural distillation strategy that transfers geometric boundary information from LiDAR to HSI by aligning their affinity distributions.
4) We validate our LSLFormer on three benchmark datasets, demonstrating its significant superiority over existing SOTA methods in both accuracy and computational efficiency (FLOPs and parameter count).

The remainder of this article is organized as follows. Section II reviews related work on HSI-LiDAR fusion. Section III describes the proposed LSLFormer architecture in detail. Section IV presents the experimental setup and analyzes the results. Finally, Section V concludes this article.

II. Related Work

In this section, we first discuss the evolution of HSI and LiDAR data fusion strategies, focusing on the two dominant deep learning paradigms: CNN- and Transformer-based models, and finally, we review the development of lightweight models in remote sensing.

A. CNN-Based Fusion Models

CNNs have been the cornerstone of HSI and LiDAR fusion for many years due to their powerful ability to extract local spatial–spectral–elevation features. In [20], LiDAR data were directly considered as another spectral band of hyperspectral data, and then, the concatenated data were fed into a CNN to learn features and perform classification. In [21], the traditional feature extraction method was combined with CNN. It fed the fused features to the CNN for learning a high-level representation and getting a classification result.

When applied to fusion, many models adopt multibranch architectures. For example, Xu et al. [22] designed a two-branch CNN (TBCNN) framework to extract features from HSI and LiDAR, respectively, and then fused them at a later stage. Hang et al. [23] integrated the HS and LiDAR data in an advanced cascade fashion by learning a coupled CNN. To better eliminate the gap between different modalities and obtain diagnostic features, Wu et al. [24] designed a cross-channel reconstruction mechanism and embedded it into a CNN-based framework. Attention mechanisms were also integrated into CNN frameworks, such as in FusAtNet [25], to selectively emphasize more informative features from each modality during the fusion process.

Despite their success, CNN-based methods are inherently constrained by their local receptive fields. They struggle to capture the global contextual relationships and long-range dependencies that are crucial for classifying complex remote sensing scenes, which can lead to classification errors, particularly in heterogeneous areas or at the boundaries of objects.

B. Transformer-Based Fusion Models

To overcome the limitations of CNNs, the Transformer architecture [12] has been introduced to the remote sensing field. Leveraging the self-attention mechanism, Transformers excel at modeling global contextual relationships. A notable example is SpectralFormer [26], which, as mentioned in our methodology, redesigns the Transformer encoder to operate on spectral sequences, effectively capturing relationships between spectral bands.

For HSI-LiDAR fusion, some researchers focused on designing a global–local fusion mechanism to integrate different modalities. For example, Ding et al. [27] introduced a global–local Transformer network (GLT-Net) to fully exploit the potential of Transformer architectures for modeling the long-range dependencies and the convolution operator, which characterizes locally correlated features in joint classification of HSI and LiDAR data. Zhang et al. [28] introduced the local information interaction Transformer (LIIT) model to capture the relationship between HSI and its complementary information (such as LiDAR), as well as to fuse the multisource features and prevent the generation of redundant information. Moreover, many researchers focused on utilizing the cross-attention mechanism for fusing the information in different remote sensing data. For example, Xue et al. [29] explored a hierarchical Transformer to fuse information from different modalities for collaborative classification of HSI and LiDAR data using cross-attention. Zhao et al. [30] integrated fractional Fourier into the Transformer to extract both global and local contextual information from HSI and LiDAR data by incorporating the cross-attention into the model.

While these Transformer-based models have set new SOTA benchmarks, their performance comes at a high cost. The self-attention mechanism has a computational complexity that is quadratic ($O(n^2)$) with respect to the sequence length. In the context of HSI, the sequence length (either spatial patches or, more critically, hundreds of spectral bands) results in prohibitive computational overhead and high memory consumption. This "computational bottleneck" hinders

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their deployment in resource-constrained environments and motivates the need for more efficient architectures.

C. Lightweight Architectures in Remote Sensing

The increasing computational demands of deep models have spurred research into lightweight and efficient model design for remote sensing tasks. A primary strategy is spectral DR. Traditional methods such as principal component analysis (PCA) [17] are often used as a preprocessing step, but this can lead to information loss. More advanced deep learning methods, such as using autoencoders (AEs) [31], have been proposed to learn a compact, nonlinear representation of the spectral data.

Another avenue is the design of efficient network architectures. This includes adopting principles from models such as MobileNet [32] (using depthwise separable convolutions) or designing compact CNNs [33]. Other techniques, such as knowledge distillation [34] and network pruning [35], have also been explored to compress large models.

However, effectively reducing the computational load of Transformers for HSI-LiDAR fusion, while simultaneously preserving crucial spectral details and facilitating effective cross-modal interaction, remains a significant challenge. Our work is situated at the intersection of Transformer-based fusion and lightweight design. Unlike methods that simply apply standard DR or generic model compression, our proposed LSLFormer introduces structurally efficient modules—H2M, MCIE, and SLA—which are specifically designed to address the unique challenges of HSI and LiDAR fusion.

III. Approach

In this section, we first review the Transformer and then describe the proposed LSLFormer in detail.

A. Preliminaries

As is well known, Transformer has not only completely changed the landscape of natural language processing but also profoundly influenced the development of computer vision. Its unique network architecture and novel scaled self-attention mechanism enable it to effectively capture global information.

1) Scaled Self-Attention: Given the transformed matrices of $Q \in \mathbb{R}^{n\times d}$, $K \in \mathbb{R}^{n\times d}$, and $V \in \mathbb{R}^{n\times d}$, the scaled dot-product attention function is formulated as follows:

$$\text{Attention}(Q,K,V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{1}$$

where n and d denote the sequential length and the feature depth, respectively.

2) Multihead Attention: To further capture the dependence between input features and improve the representational ability, (1) was extended to a multihead version denoted as multihead attention (MHA), which consists of h paralleled "heads" to make the scale dot-product attention. Specifically, given the input features $Q \in \mathbb{R}^{n\times d}$, MHA is formulated as follows:

$$\text{MHA}(X) = [\text{head}_1, \ldots, \text{head}_h]$$

where

$$\text{head}_i = \text{Attention}\left(Q^i, K^i, V^i\right) \tag{2}$$

where Q^i, K^i, and V^i are the split features for the ith head, and they are obtained by truncating the input features X after projection. Meanwhile, a linear-merge projection is used after MHA, defined as follows:

$$O = \text{MHA}(X)\,W_O \tag{3}$$

where $W_O \in \mathbb{R}^{d\times d}$ is the weight matrix and $O \in \mathbb{R}^{n\times d}$ is the final output.

3) Feed-Forward Network: To enhance the feature extraction capability of the model, a feed-forward network (FFN) makes a positionwise and dense transformation from the input features to the output ones, which keeps the nonlinearity by expanding the hidden units in the ReLU layer. The formulation of FFN can be written as follows:

$$\text{FFN}(X) = \max(0, XW_1 + b_1)\,W_2 + b_2 \tag{4}$$

where $W_1 \in \mathbb{R}^{d\times d_f}$, $W_2 \in \mathbb{R}^{d_f\times d}$, $b_1 \in \mathbb{R}^{d_f}$ and $b_2 \in \mathbb{R}^d$ are weights and bias, and $\max(\cdot)$ is the ReLU activation.

B. Lightweight Spectral–LiDAR Former

The proposed LSLFormer aims to achieve efficient HSI and LiDAR fusion and high-precision classification of remote sensing images with fewer parameters and lower computational cost. As shown in Fig. 1, the LSLFormer adopts the ViT as its backbone but only has three layers of encoders. Instead of the traditional ViT approach that models image patch sequences, LSLFormer prioritizes spectral characteristics by modeling the spectral sequence like SpectralFormer [26].

The fusion mechanism incorporates two distinct interaction stages between the HSI and LiDAR modalities. The first occurs at the embedding stage: the HSI, after DR via the H2M module, is jointly embedded with the LiDAR data that have been preprocessed by the MCIE module. The second interaction is implemented via a cross-modal attention mechanism: the features extracted from the LiDAR data by the MCIE module are introduced into the SLA module along with the HSI spectral features to compute attention weights. Furthermore, acknowledging the proven effectiveness of the groupwise spectral embedding (GSE) initially proposed in SpectralFormer [26], we have integrated this technique into our network architecture.

1) Hyperspectral-to-Multispectral: Conventional spectral DR methods often ignore the sequential nature of spectral bands or fail to align with the downstream classification objective. To mitigate this, we introduce the H2M module that draws inspiration from the optical physics of sensor spectral integration. In remote sensing, a multispectral sensor acquires data by integrating the continuous spectral radiance $R(\lambda)$ weighted by a specific SRF [18]. For the ith multispectral band, this process is formulated as follows:

$$M_i = \frac{\int R(\lambda)\,\text{SRF}_i(\lambda)\,d\lambda}{\int \text{SRF}_i(\lambda)\,d\lambda} \tag{5}$$

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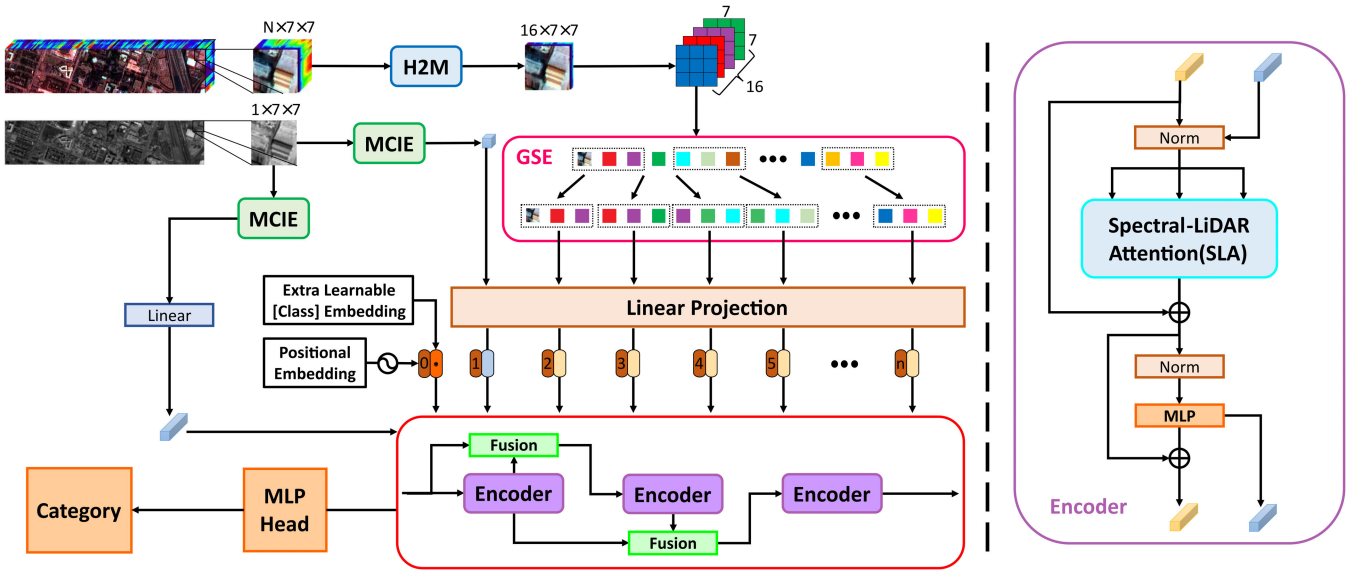


Fig. 1. Overview illustration of the proposed LSLFormer for remote sensing image classification task. To alleviate the high computational burden of the self-attention mechanism in spectral modeling, we propose an H2M module that performs physically motivated spectral downsampling. Moreover, to achieve effective and complementary fusion between the spectral information of HSI and the elevation information provided by LiDAR data, we use a multiscale channel interaction and enhancement (MCIE) module to enhance LiDAR feature representation and introduce an SLA mechanism.

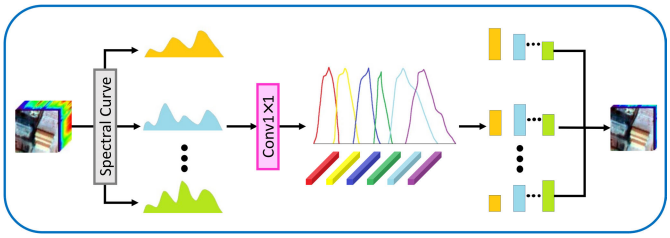


Fig. 2. Schematic of the H2M module.

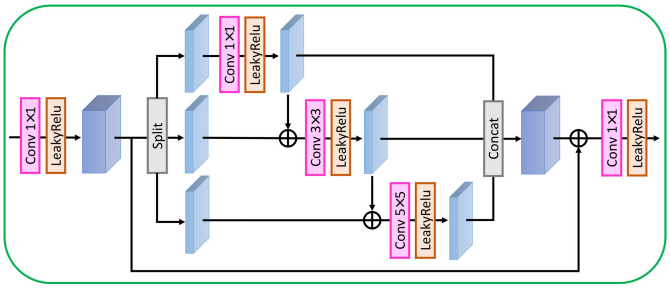


Fig. 3. Detailed structure of the multiscale channel interaction enhancement module.

where M_i denotes the response of the ith multispectral band. Translating this physical process into the discrete domain of hyperspectral data, we can view the generation of a reduced feature map as a weighted summation of the original high-dimensional bands. We formulate this as a learnable spectral integration process

$$\hat{M}_i = \frac{\sum_{b=1}^{B_h} \theta_{\mathrm{SRF}}^{(i,b)} \cdot H_b}{\sum_{b=1}^{B_h} \theta_{\mathrm{SRF}}^{(i,b)}} \tag{6}$$

where H_b denotes the reflectance at the bth hyperspectral band and the learnable kernel parameter $\theta_{\mathrm{SRF}}^{(i,b)}$ corresponds to the SRF weight between the bth hyperspectral and the ith multispectral band.

To ensure that this operation mimics the physical properties of spectral integration (where response values are nonnegative and normalized), we implement (6) using a constrained pointwise (1×1) convolution, as shown in Fig. 2. Specifically, we apply a normalization operation to the convolution kernels to enforce the physical constraints

$$\hat{M} = \mathrm{Clamp}\left(\mathrm{Conv}_{1\times1}\left(H;\, \sigma\left(\theta_{\mathrm{SRF}}\right)\right), [0, 1]\right) \tag{7}$$

where $H \in \mathbb{R}^{B\times C\times N}$ is the hyperspectral patch, $\hat{M} \in \mathbb{R}^{B\times 16\times N}$ is the network generated multispectral representation, and $\sigma(\cdot)$ represents a normalization function. Unlike fixed physical sensors, the "virtual SRFs" in H2M are dynamically optimized during training. This allows the network to learn the optimal spectral aggregation strategy specifically for distinguishing land cover materials in the given dataset, effectively bridging the gap between high-dimensional redundancy and computational efficiency.

2) Multiscale Channel Interaction Enhancement: Before entering the encoder, LiDAR features are processed by a MCIE module that emphasizes spatial structure extraction. The module is depicted in Fig. 3. Let the input be $L \in \mathbb{R}^{C_{\mathrm{in}}\times N}$. A pointwise (1×1) convolutional layer with LeakyReLU [36] produces a compact intermediate representation $L' \in \mathbb{R}^{C_m \times N}$

$$L' = \sigma\left(\mathrm{Conv}_{1\times1}\left(L\right)\right) \tag{8}$$

where C_m is a modest channel budget (set empirically). L' is evenly split into three channel groups

$$L' = [L_1, L_2, L_3]\,. \tag{9}$$

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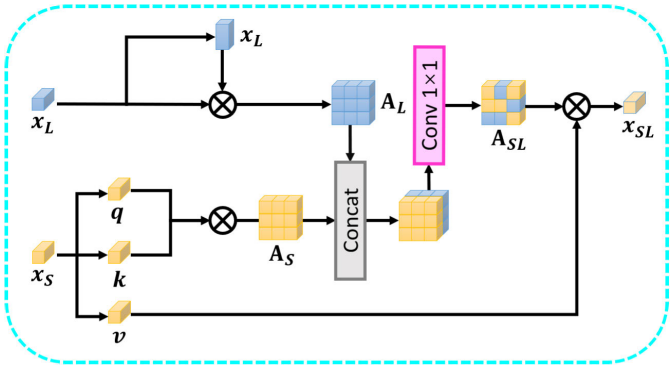


Fig. 4. Detailed structure of SLA.

Each group is processed by a branch with a different spatial receptive field (kernels $k_1 = 1$, $k_2 = 3$, and $k_3 = 5$), and branches are connected progressively to enable information propagation

$$Y_1 = \sigma\left(g\left(\text{Conv}_{k_1}(L_1)\right)\right) \tag{10}$$

$$Y_2 = \sigma\left(g\left(\text{Conv}_{k_2}(L_2 + Y_1)\right)\right) \tag{11}$$

$$Y_3 = \sigma\left(g\left(\text{Conv}_{k_3}(L_3 + Y_2)\right)\right) \tag{12}$$

where $\sigma(\cdot)$ denotes LeakyReLU and $g(\cdot)$ denotes GroupNorm [37]. The branch outputs are concatenated and fused with a residual connection to preserve the original information

$$Z = [Y_1, Y_2, Y_3], \quad O = \sigma\left(\text{Conv}_{1\times1}\left(L' + Z\right)\right) \tag{13}$$

where $O \in \mathbb{R}^{C_{\text{out}} \times N}$ is the final output of MCIE module. By focusing on multiscale spatial convolutions, MCIE extracts complementary geometric cues from replicated LiDAR channels with minimal overhead.

3) Spectral–LiDAR Attention: Standard spectral self-attention solely relies on spectral similarities, potentially confusing materials with similar spectra but distinct spatial geometries. To address this, we propose the SLA module to explicitly incorporate LiDAR-derived structural guidance into the attention mechanism. As shown in Fig. 4, the SLA module operates on two parallel streams. First, we compute the standard spectral self-attention map $A_S \in \mathbb{R}^{n\times n}$, which captures the long-range dependencies between spectral bands

$$A_S = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \tag{14}$$

where $Q, K \in \mathbb{R}^{n\times d}$ are query and key projections of the spectral features. Simultaneously, we construct a structural affinity matrix A_L using the LiDAR features extracted by the MCIE module. We formulate A_L as a normalized probability distribution to ensure mathematical compatibility with the spectral attention

$$A_L = \text{Softmax}\left(\frac{X_L X_L^T}{\sqrt{d}}\right) \tag{15}$$

where $X_L \in \mathbb{R}^{n\times d}$ represents the transformed LiDAR features. Here, $A_L(i, j)$ quantifies the geometric connectivity between token i and token j. To fuse these two complementary attention distributions, we concatenate A_S and A_L along a new dimension and apply a lightweight convolution to learn the optimal mixing strategy

$$A_{SL} = \text{Conv}_{1\times1}\left(\text{Concat}\left([A_S, A_L]\right)\right) \tag{16}$$

where $\text{Conv}_{1\times1}$ maps the two-channel input to a single unified attention map. This operation effectively acts as a learnable gate, allowing the network to dynamically weigh whether to trust the spectral correlation or the structural continuity for each token pair. Finally, the fused attention map is applied to the value projection $V \in \mathbb{R}^{n\times d}$ of the spectral features

$$X_{SL} = A_{SL} \cdot V. \tag{17}$$

This ensures that the rich spectral information serves as the fundamental feature basis, while the attention weights are refined by the fused structural–spectral guidance.

4) Cross-Modal Structural Consistency: Direct regression between spectral features and LiDAR elevation is physically ill-posed due to the domain gap. To address this, we propose the CMSC loss to align the underlying spatial topology rather than raw values. We hypothesize that pixels with high geometric affinity in LiDAR data should exhibit similar spectral correlations. Specifically, for a flattened feature map $X \in \mathbb{R}^{C\times N}$, we construct a pairwise affinity distribution to represent its structural topology. The probability of connection between pixel i and j is defined using a temperature-scaled cosine similarity

$$P_{i,j} = \text{Softmax}_j\left(\frac{X_i \cdot X_j}{\tau \|X_i\| \ \|Xj\|}\right) \tag{18}$$

where τ controls the distribution sharpness. We compute this distribution for both the LiDAR data (P^L, serving as the structural teacher) and the HSI features (Q^H). To transfer geometric knowledge, we minimize the Kullback–Leibler (KL) divergence between them

$$\mathcal{L}_{\text{CMSC}} = \frac{1}{N}\sum_{i=1}^{N} D\text{KL}\left(P_i^L \middle|\middle| Q_i^H\right). \tag{19}$$

The final objective function combines the classification loss and the structural regularization

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda\mathcal{L}_{\text{CMSC}} \tag{20}$$

where λ balances the contribution of structural alignment (set to 0.01 empirically).

IV. Experiments

In this section, we consider three different HSIs and their associated LiDAR data to evaluate the performance of the proposed spectral–LiDAR fusion network.

A. Dataset Description

1) Houston 2013: This dataset was acquired by the compact airborne spectrographic imager (CASI), covering the University of Houston campus and its surrounding urban area [38]. The scene comprises a total of 349 × 1905 pixels. The HSI component contains 144 spectral bands, spanning the

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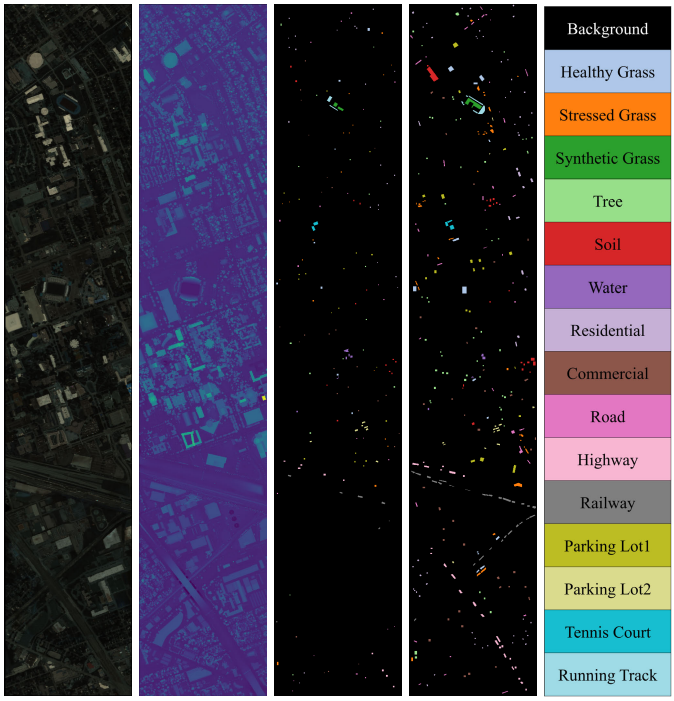


Fig. 5. Houston 2013 data. (a) Pseudocolor image from the HSI data using bands 70, 50, and 30, respectively. (b) LiDAR image. (c) Ground truth of the training set. (d) Ground truth of the disjoint testing set. (e) Legend.

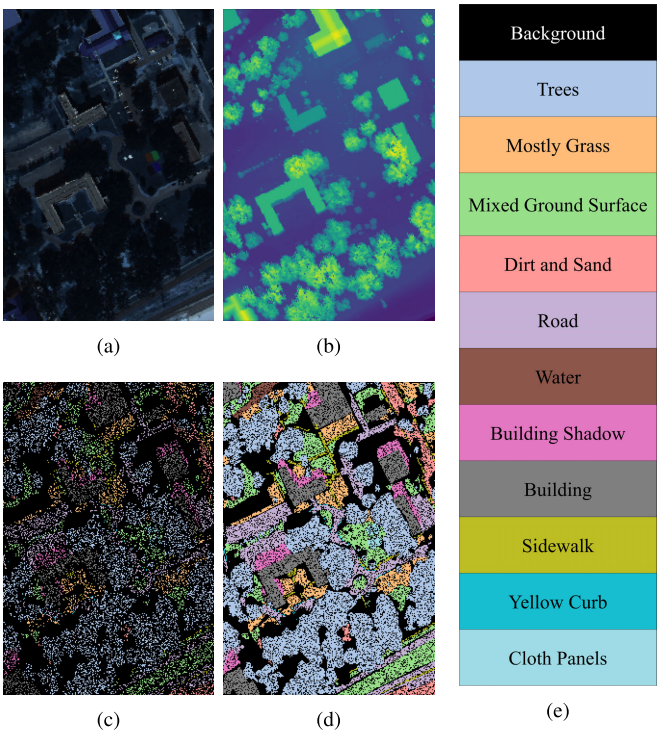


Fig. 6. MUUFL data. (a) Pseudocolor image from the HSI data using bands 21, 11, and 1, respectively. (b) LiDAR image. (c) Ground truth of the disjoint training set. (d) Ground truth of the testing set. (e) Legend.

TABLE I

DESCRIPTION OF THE HOUSTON 2013 DATASET, INCLUDING THE TYPES OF GROUND OBJECTS AND THE CORRESPONDING NUMBERS OF TRAINING AND TESTING SAMPLES

Class No.	Class Name	Training Set	Testing Set
1	Healthy Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot1	192	1041
13	Parking Lot2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
	Total	2832	12197

TABLE II

DESCRIPTION OF THE MUUFL DATASET, INCLUDING THE TYPES OF GROUND OBJECTS AND THE CORRESPONDING NUMBERS OF TRAINING AND TESTING SAMPLES

Class No.	Class Name	Training Set	Testing Set
1	Trees	5811	17435
2	Mostly Grass	1067	3203
3	Mixed Ground Surface	1720	5162
4	Dirt and Sand	456	1370
5	Road	1671	5016
6	Water	116	350
7	Building Shadow	558	1675
8	Building	1560	4680
9	Sidewalk	346	1039
10	Yellow Curb	45	138
11	Cloth Panels	67	202
	Total	13417	40270

380–1050-nm region with a spectral resolution of 10 nm. The coregistered LiDAR data consist of a single channel recording elevation information. The ground truth encompasses 15 distinct land cover and land use classes, as illustrated in Fig. 5. The division into training and test sets adheres to the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest. Detailed statistics, including class names and sample sizes, are provided in Table I.

2) MUUFL: This dataset scene covers the University of Southern Mississippi campus, comprising a total of 325×220 pixels [39]. The HSI component was acquired by a reflective optics system imaging spectrometer (ROSIS) sensor, initially containing 72 spectral bands spanning the wavelength range of 375–1050 nm. After the removal of bands severely affected by water absorption, the number of bands was reduced to 64. The LiDAR data were collected by the Gemini ALTM LiDAR sensor and consist of two bands. The ground truth encompasses 11 distinct land cover and land use classes, as shown in Fig. 6. Detailed data statistics, based on a random selection of 25% samples from each class used for training, are provided in Table II.

3) Trento: The dataset scene covers a rural area south of Trento, Italy, and consists of 600×166 pixels [40]. The HSI

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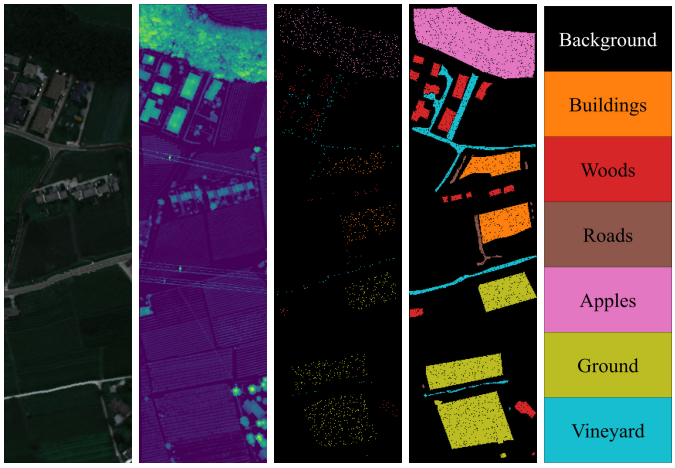


Fig. 7. Trento data. (a) Pseudocolor image from the HSI data using bands 20, 15, and 10, respectively. (b) LiDAR image. (c) Ground truth of the disjoint training set. (d) Ground truth of the testing set. (e) Legend.

TABLE III
DESCRIPTION OF THE TRENTO DATASET, INCLUDING THE TYPES OF GROUND OBJECTS AND THE CORRESPONDING NUMBERS OF TRAINING AND TESTING SAMPLES

Class No.	Class Name	Training Set	Testing Set
1	Buildings	201	3833
2	Woods	145	2758
3	Roads	23	456
4	Apples	456	8667
5	Ground	525	9976
6	Vineyard	158	3016
	Total	1508	28706

data were acquired by the AISA Eagle sensor, encompassing 63 bands with a wavelength range of 420–990 nm and a spectral resolution of 9.2 nm. The LiDAR data were collected by the Optech ALTM 3100EA sensor, providing a single channel for elevation information. The scene features six mutually exclusive land cover classes, as depicted in Fig. 7. Detailed statistics, including class names and sample sizes, are provided in Table III.

B. Experimental Setup

1) Comparison With SOTA Backbone Networks: In order to study the effectiveness of the proposed spectral–LiDAR fusion network, we have performed extensive experiments and compared it with several representative convolutional networks, i.e., 1D-CNNs [41], 2D-CNNs [9], CCR-Net [24], and SOTA Transformer networks, i.e., ViT [13], SpectralFormer [26], FrIT [30], S2ATNet [42], and MobileViTv2 [43].

2) Implementation Details: All experiments were conducted on a workstation equipped with an Intel i5-14600KF CPU, 32 GB of RAM, and an NVIDIA RTX 4060 8-GB GPU. The testing environment utilized PyTorch 2.5.1 and Python 3.10.16. For training and evaluating model performance, a batch size of 64 was employed, and image patches of size 7×7 were extracted from the hyperspectral and LiDAR images without any data augmentation strategy. All models were trained using the Adam optimizer with the learning rate initialized to 5×10^{-4} and the weight decay set to 5×10^{-3}. The number of training epochs varied depending on the dataset: 600 epochs for Houston 2013, 500 epochs for MUUFL, and 200 epochs for Trento. Furthermore, a step scheduler was applied, with the step size set to one-tenth of the total epochs and a gamma value of 0.9. During training, validation was performed every 10 epochs, and the highest achieved value is reported for each experiment.

3) Evaluation Matrix: The performance of the proposed network is evaluated using several standard quantitative metrics and compared against other methods, specifically overall accuracy (OA), average accuracy (AA), and the statistical kappa coefficient (κ).

C. Quantitative Analysis and Visual Evaluation

Quantitative classification results, summarized by the three overall performance indices, i.e., OA, AA, and κ, along with the per-class accuracies, are reported in Tables IV–VI for the Houston 2013, MUUFL, and Trento datasets, respectively. The highest accuracy value in each category is highlighted in bold. A qualitative evaluation is further performed by visualizing the classification maps generated by the different methods. The resultant classification maps for the Houston 2013, MUUFL, and Trento datasets are presented in Figs. 8–10, respectively.

1) Analysis on the Houston 2013 Dataset: Table IV displays the detailed classification results for all methods on the Houston 2013 dataset. LSLFormer achieved the best performance across all three overall metrics, attaining an OA of 92.15%, an AA of 92.92%, and a κ coefficient of 91.48%.

Compared to the convolutional networks, LSLFormer's OA was 4.51% higher than the best-performing CCR-Net (87.64%). Among the Transformer-based networks, LSLFormer also demonstrated a significant advantage, with its OA surpassing FrIT (88.90%), S2ATNet (87.51%), and MobileViTv2 (86.62%). Out of the 15 classes, LSLFormer only achieved the highest accuracy in two categories (i.e., Classes 8 and 13), but its performance in all categories was very balanced. Although FrIT and CCR-Net performed exceptionally well on certain specific classes (e.g., FrIT on Classes 3, 7, 10, and 14), LSLFormer's comprehensive lead in overall metrics demonstrates its superior generalization ability and stability in handling complex, high-dimensional hyperspectral data. In Fig. 8, it can be clearly observed that LSLFormer makes precise delineation for Commercial (brown) and Parking Lot2 (khaki) areas. Compared to other models, LSLFormer has clearer category boundaries and correctly distinguishes the location information of roads and green spaces around commercial streets.

2) Analysis on the MUUFL Dataset: Table V presents the comparative results on the MUUFL dataset. LSLFormer once again led in the key metrics, obtaining a 94.40% OA, an 87.57% AA, and a 92.62% κ coefficient.

It is noteworthy that on this dataset, FrIT achieved the highest AA (87.86%), slightly surpassing LSLFormer (87.57%), which suggests that FrIT performed well in balancing different classes (especially those with fewer samples). However,

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TABLE IV

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND KAPPA (IN %), AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE HOUSTON 2013 DATASET. THE BEST ONE IS SHOWN IN BOLD

Class No.	Convolutional Networks			Transformer Networks					
	1D-CNNs	2D-CNNs	CCR-Net	ViT	SpectralFormer	FrIT	S2ATNet	MobileViTv2	LSLFormer
1	97.43	92.11	**100.00**	97.76	83.95	98.88	98.52	**100.00**	99.12
2	87.24	92.53	84.00	84.17	**99.44**	93.27	82.90	72.73	93.95
3	93.52	99.60	94.54	99.19	94.06	**100.00**	96.18	91.11	99.60
4	96.43	92.25	92.65	97.54	**97.92**	92.23	96.09	96.31	95.72
5	91.19	86.81	**100.00**	96.70	99.43	97.06	95.05	96.61	99.06
6	82.42	98.52	92.52	84.94	96.50	90.13	85.66	**100.00**	97.95
7	82.39	90.07	90.10	78.90	90.02	**94.19**	85.67	89.97	89.80
8	87.54	70.17	78.01	84.52	81.01	78.98	86.16	79.43	**91.53**
9	60.49	77.91	**89.14**	73.16	79.60	77.37	83.97	88.86	82.13
10	89.17	84.53	77.88	77.02	52.41	**92.92**	88.02	83.36	82.79
11	**86.71**	74.69	68.65	61.83	86.62	73.74	72.23	80.62	85.10
12	88.76	89.86	**98.16**	81.47	83.29	92.60	84.46	78.10	92.57
13	67.16	84.72	80.57	87.73	74.39	74.72	80.82	93.93	**96.37**
14	87.86	95.72	88.64	62.53	97.17	**100.00**	88.49	86.52	96.36
15	**100.00**	96.03	**100.00**	94.95	97.25	95.93	95.54	99.77	99.79
OA	85.72	86.07	87.64	83.39	86.37	88.90	87.51	86.62	**92.15**
AA	87.51	88.09	88.85	86.09	87.54	90.84	89.11	87.76	**92.92**
$\kappa(\times 100)$	84.50	86.58	78.15	85.20	79.26	87.95	86.45	85.47	**91.48**

TABLE V

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND KAPPA (IN %), AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE MUUFL DATASET. THE BEST ONE IS SHOWN IN BOLD

Class No.	Convolutional Networks			Transformer Networks					
	1D-CNNs	2D-CNNs	CCR-Net	ViT	SpectralFormer	FrIT	S2ATNet	MobileViTv2	LSLFormer
1	96.80	96.08	93.40	94.26	97.98	97.09	96.93	93.61	**99.12**
2	79.52	81.73	76.81	80.45	71.40	81.09	78.25	78.39	**93.95**
3	85.20	83.23	71.79	82.85	82.08	84.53	87.60	**93.81**	90.90
4	94.04	93.29	62.98	93.61	80.51	94.31	89.93	95.34	**96.42**
5	89.89	92.72	84.37	92.99	89.33	94.17	95.01	95.96	**96.23**
6	**99.32**	96.54	83.60	93.39	89.43	97.69	91.84	92.93	94.29
7	78.58	80.87	84.34	89.34	81.13	76.97	81.92	**91.12**	86.69
8	94.33	96.06	89.79	94.88	94.68	96.31	96.11	90.16	**97.54**
9	79.93	**83.01**	80.00	68.26	75.07	82.18	78.94	50.51	74.88
10	**96.72**	95.77	0	50.00	23.19	81.90	81.82	0	29.71
11	97.79	**99.45**	94.20	96.52	77.72	92.82	95.10	98.39	88.12
OA	91.55	91.87	85.99	90.94	90.05	92.35	92.51	90.23	**94.40**
AA	81.67	84.09	66.18	76.00	78.41	**87.86**	86.00	78.48	87.57
$\kappa(\times 100)$	88.83	89.23	87.90	92.19	86.78	89.90	90.10	86.97	**92.62**

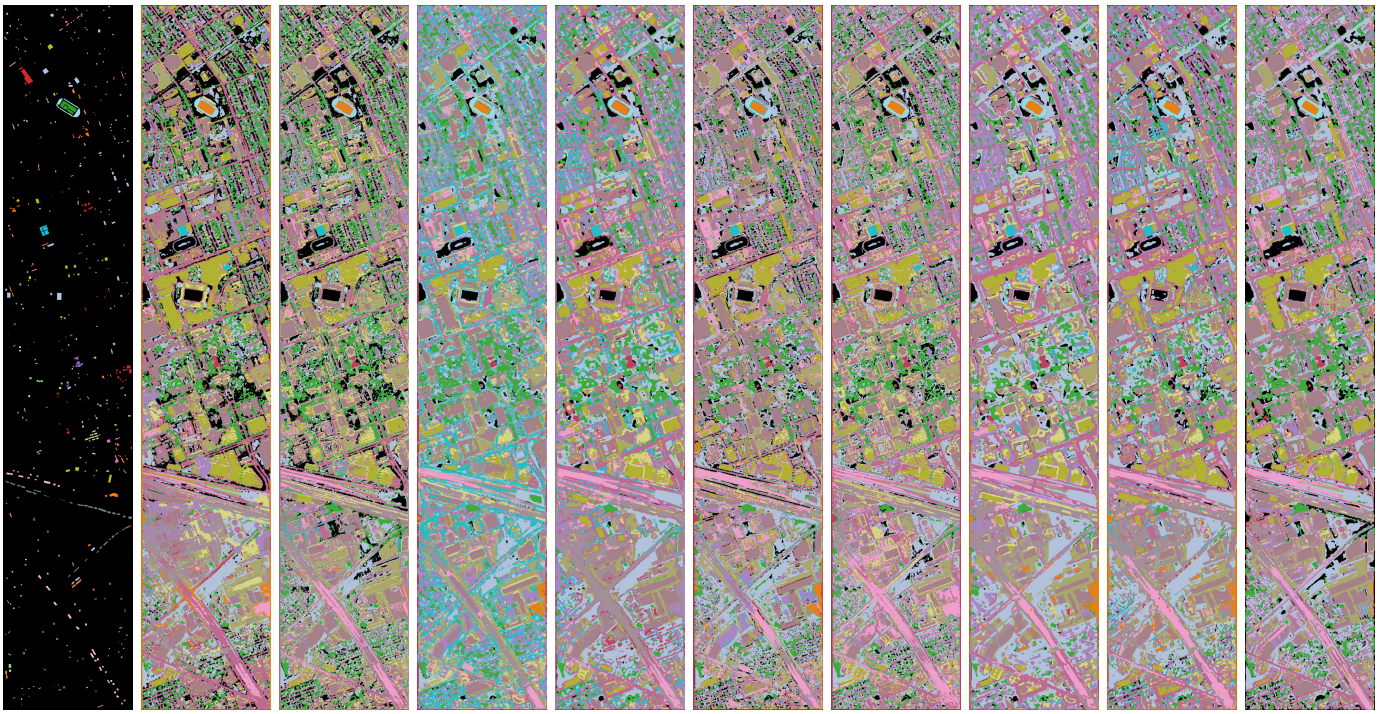
TABLE VI

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND KAPPA (IN %), AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE TRENTO DATASET. THE BEST ONE IS SHOWN IN BOLD

Class No.	Convolutional Networks			Transformer Networks					
	1D-CNNs	2D-CNNs	CCR-Net	ViT	SpectralFormer	FrIT	S2ATNet	MobileViTv2	LSLFormer
1	94.27	97.62	94.41	93.44	97.31	98.82	**99.77**	98.60	99.37
2	84.51	86.70	95.26	83.98	96.59	84.67	89.97	83.91	**98.29**
3	99.76	97.91	62.21	88.00	78.07	97.96	98.45	98.00	**100.00**
4	99.61	99.53	**100.00**	**100.00**	99.76	99.26	99.91	99.63	**100.00**
5	99.41	99.06	99.76	98.42	90.27	98.94	**99.77**	98.88	99.42
6	93.38	88.10	**99.55**	77.11	96.39	85.59	95.56	90.12	98.02
OA	96.60	96.66	97.81	94.35	95.13	96.24	98.38	96.66	**99.34**
AA	94.23	94.00	94.66	86.46	93.06	93.61	97.11	94.62	**98.45**
$\kappa(\times 100)$	95.48	95.54	97.09	92.47	93.56	94.99	97.83	95.54	**99.12**

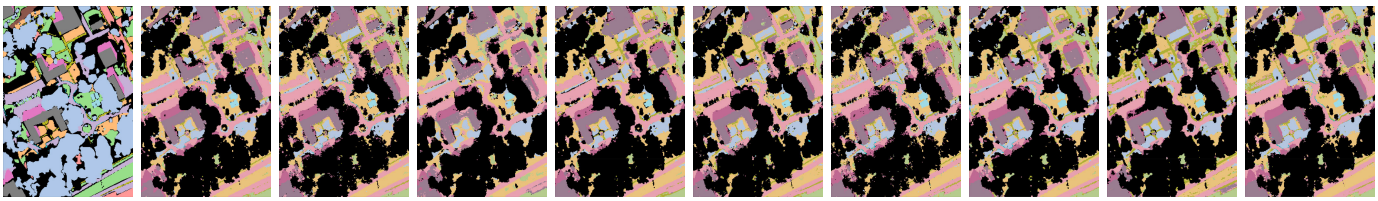
LSLFormer's significant advantage in OA and κ (e.g., 2.05% higher OA than FrIT) indicates its greater accuracy across the dataset as a whole. S2ATNet also delivered a strong performance on this dataset, achieving the second-highest

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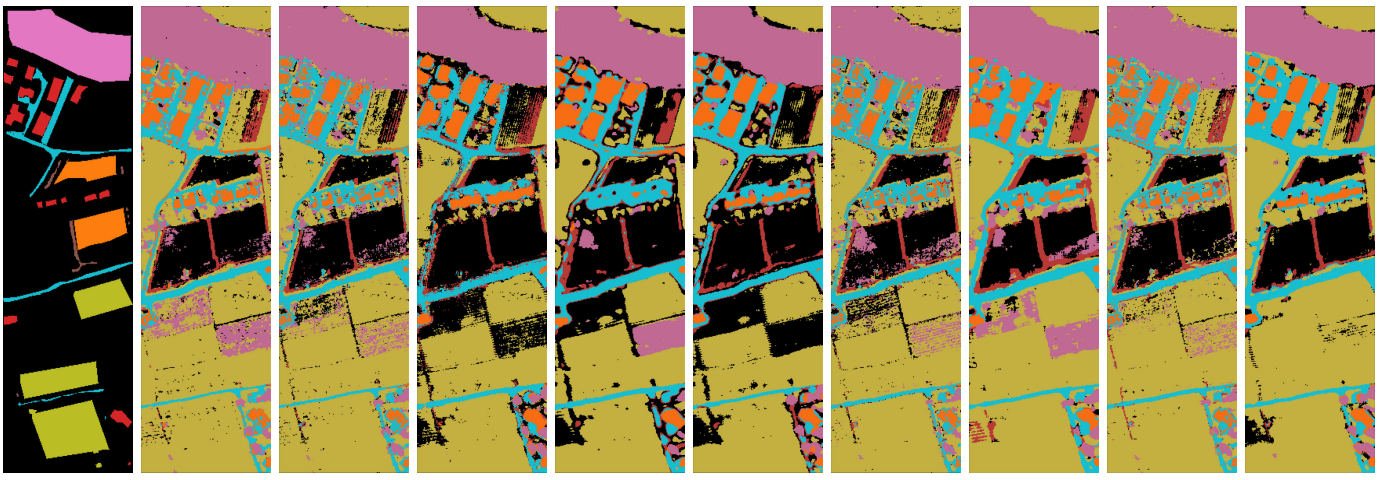
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 8. Classification maps for the Houston 2013 dataset. (a) Ground truth. (b) 1D-CNNs (85.72%). (c) 2D-CNNs (86.07%). (d) CCR-Net (87.64%). (e) ViT (83.39%). (f) SpectralFormer (86.37%). (g) FrIT (88.90%). (h) S2ATNet (87.51%). (i) MobileViTv2 (86.62%). (j) LSLFormer (92.15%).



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. Classification maps for the MUUFL dataset. (a) Ground Truth. (b) 1D-CNNs (91.55%). (c) 2D-CNNs (91.87%). (d) CCR-Net (85.99%). (e) ViT (90.94%). (f) SpectralFormer (90.05%). (g) FrIT (92.35%). (h) S2ATNet (92.51%). (i) MobileViTv2 (90.23). (j) LSLFormer (94.40%).



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 10. Classification maps for the Trento dataset. (a) Ground Truth. (b) 1D-CNNs (96.60%). (c) 2D-CNNs (96.66%). (d) CCR-Net (97.81%). (e) ViT (94.35%). (f) SpectralFormer (95.13%). (g) FrIT (96.24%). (h) S2ATNet (98.38%). (i) MobileViTv2 (96.66%). (j) LSLFormer (99.34%).

OA (92.51%) and κ coefficient (90.10%). As shown in Fig. 9, LSLFormer provides more precise boundary delineation for Mostly Grass (orange), Road (light purple), and Building (gray), while boundaries between adjacent categories are clearer, and confusion phenomena are significantly reduced.

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TABLE VII

ABLATION ANALYSIS OF THE PROPOSED LSLFORMER WITH A COMBINATION OF DIFFERENT MODULES ON THE HOUSTON 2013 DATASET. THE BEST ONE IS SHOWN IN BOLD

Ablation Settings				Houston 2013		
H2M	MCIE	SLA	CMSC	OA(%)	AA(%)	κ
×	×	×	×	82.85	83.51	81.37
×	✓	✓	✓	87.19	87.28	86.09
✓	×	✓	✓	87.48	88.35	86.40
✓	✓	×	✓	87.63	88.74	86.56
✓	✓	✓	×	88.64	89.37	87.66
✓	✓	✓	✓	**92.15**	**92.92**	**91.48**

3) Analysis on the Trento Dataset: The results in Table VI are from the Trento multimodal dataset. On this dataset, LSLFormer showed overwhelming superiority, achieving the top rank in all three metrics: OA, AA, and κ, with scores of 99.34%, 98.45%, and 99.12%, respectively.

These near-perfect scores are significantly better than all other comparative methods. For instance, LSLFormer's κ coefficient was 1.29% higher than the second-place S2ATNet (97.83%) and 4.13% higher than FrIT (94.99%). Among the six classes, LSLFormer significantly outperformed other methods on Class 2 (98.29%) and maintained exceptionally high accuracy on all other classes (all exceeding 98%), showcasing its powerful feature extraction and classification capabilities. In Fig. 10, it can be clearly observed that LSLFormer demonstrates robust classification performance across all categories, particularly achieving significantly higher classification accuracy for the Woods (red) compared to the other competitive methods.

Synthesizing the quantitative data from all three tables, LSLFormer consistently surpassed all comparative convolutional networks (1D-CNNs, 2D-CNNs, and CCR-Net) and Transformer networks (ViT, SpectralFormer, FrIT, S2ATNet, and MobileViTv2) in terms of OA and the Kappa coefficient across all three datasets. These experimental results strongly validate the effectiveness and competitiveness of the LSLFormer architecture in the remote sensing image classification task.

D. Model Analysis

1) Ablation Study: To systematically evaluate the effectiveness of the proposed LSLFormer network architecture, extensive ablation experiments were conducted on the Houston 2013 dataset, focusing on four critical components: H2M, MCIE, SLA, and CMSC. The results of these experiments are summarized in Table VII.

Specifically, the baseline model exhibited the poorest classification performance. The OA, AA, and κ coefficients dropped by 9.3%, 9.41%, and 10.11%, respectively, compared to the complete LSLFormer model. This observation, to some extent, suggests that a naive application of the ViT architecture for fusing spectral and LiDAR data, without sufficient extraction and utilization of the elevation information from the LiDAR data, is suboptimal for the remote sensing image classification task. Removing any single module from the complete LSLFormer model resulted in a degradation of performance. This confirms that every module included in the ablation study contributes positively to enhancing the model's classification accuracy. Furthermore, we observe that the H2M module provides the most significant contribution to the model's performance improvement.

TABLE VIII

ANALYSIS OF DIFFERENT SPECTRAL DOWNSAMPLING TECHNIQUES ON THE HOUSTON 2013 DATASET. THE BEST ONE IS SHOWN IN BOLD

Method	Houston 2013		
	OA(%)	AA(%)	κ
PCA	8.98	79.04	77.21
SPA	86.40	87.36	85.23
Cor	87.68	87.52	86.61
K-means	87.72	88.72	86.66
H2M	**92.15**	**92.92**	**91.48**

2) Different Spectral Downsampling Techniques: To validate the superiority of the proposed H2M, we conduct a comparative analysis against several conventional hyperspectral data spectral downsampling techniques on the Houston 2013 dataset. These methods include PCA [17], successive projection algorithm (SPA) [44], correlation-based clustering (Cor), and the K-means clustering (K-means) [45]. For a fair comparison, all methods reduced the number of HSI bands in the Houston 2013 dataset from 144 to 16.

As shown in Table VIII, the proposed H2M module achieves the best performance across all three metrics, securing the highest OA of 92.15%, AA of 92.92%, and κ of 91.48. This demonstrates a significant margin of improvement over the traditional spectral downsampling methods. The second-best performing method is K-means, which yielded an OA of 87.72%, an AA of 88.72%, and a κ of 86.66. Our H2M module surpasses K-means by a notable 4.43% in OA, 4.2% in AA, and 4.82 in κ, respectively. The other techniques, Cor (87.68% OA) and SPA (86.40% OA), show moderate results, while PCA achieves the lowest performance with an OA of 78.98%. This comparison effectively demonstrates that the H2M module is more effective at spectral downsampling compared to conventional statistical and clustering-based approaches.

3) Computational Cost Analysis: To comprehensively evaluate the efficiency of the proposed LSLFormer model, we conducted a visual comparative analysis of its computational cost against existing baseline methods on the Houston 2013 dataset. Fig. 11 illustrates the distribution of various models across three key dimensions: OA, %, computational complexity (FLOPs, M), and the number of model parameters (Params, K). An ideal model would be positioned in the upper-left quadrant, signifying the achievement of maximum classification accuracy (high OA) with minimal computational overhead (low FLOPs).

From Fig. 11, it can be observed that traditional convolution-based models (e.g., 1D-CNNs and 2D-CNNs) achieve moderate accuracy (85%–86%) with relatively low computational costs. The introduction of spectral–spatial coupling in CCR-Net slightly increases both FLOPs and accuracy, indicating the benefit of incorporating contextual dependencies. Transformer-based architectures (e.g., SpectralFormer,

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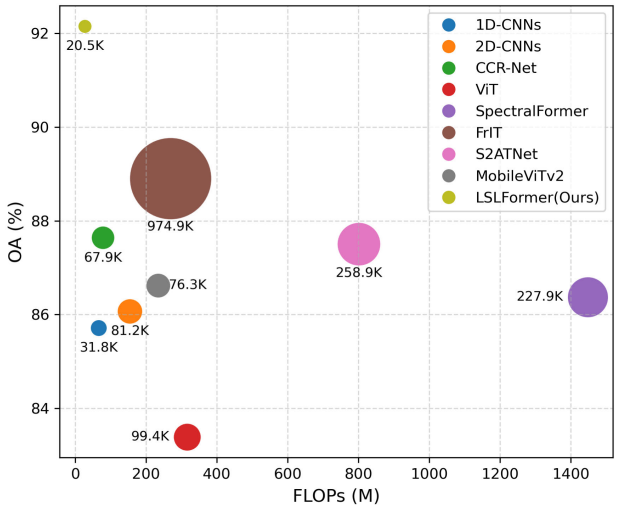


Fig. 11. Computation cost and OA of different methods (i.e., 1D-CNNs, 2D-CNNs, CCR-Net, ViT, SpectralFormer, FrIT, S2ATNet, MobileViTv2, and the proposed LSLFormer) on the Houston 2013 dataset. The vertical axis represents OA (%), the horizontal axis denotes FLOPs (M), and the size of each data point is directly proportional to the model's parameter count (Params, K).

TABLE IX
MODULE-LEVEL COMPUTATIONAL COMPLEXITY ANALYSIS ON THE HOUSTON 2013 DATASET

Module	Params (K)	Params (%)	FLOPs (M)	FLOPs (%)
H2M	2.30	11.2	7.23	26.9
MCIE	0.73	3.5	1.54	5.8
SLA	14.00	68.3	13.43	50.1
Others	3.48	17.0	4.62	17.2
Total	**20.51**	**100**	**26.82**	**100**

FrIT, and S2ATNet) generally demonstrate higher model complexity, reflected by significantly increased FLOPs and parameter counts. It is worth noting that by introducing separable self-attention, the MobileViTv2 significantly reduces the quadratic computational complexity compared to other Transformer-based architectures.

By contrast, our proposed LSLFormer model achieves the optimal balance among all compared methods. LSLFormer is positioned in the upper-left quadrant, attaining the highest OA of 92.15% while maintaining exceptionally low computational complexity. Specifically, LSLFormer requires only 26.82M FLOPs and 20.51k parameters. When compared to the runner-up model, FrIT, LSLFormer demonstrates a 3.25% point improvement in OA (92.15% versus 88.9%). Concurrently, its computational cost is significantly lower, utilizing only approximately 10% of the FLOPs (26.82/268.59) and less than 3% of the parameters (20.51/974.92) required by FrIT.

Table IX reports the module-level computational complexity of the proposed LSLFormer. It can be observed that the SLA module accounts for the largest proportion of computational resources (about 50% of FLOPs). This is expected, as SLA serves as the core feature extractor responsible for complex global spectral–spatial modeling. Crucially, however, the absolute cost of SLA is kept remarkably low. This is directly attributed to the preceding H2M module, which aggressively compresses the high-dimensional spectral sequence. Meanwhile, the H2M and MCIE modules themselves are extremely lightweight (contributing only about 11% and 3% of parameters, respectively).

In summary, this analysis compellingly demonstrates the superiority of the LSLFormer architecture. It significantly enhances classification accuracy while drastically reducing computational and storage requirements, establishing it as a highly efficient and competitive solution.

V. CONCLUSION

In this article, we propose a new LSLFormer to fuse HSI and LiDAR data for land cover classification. The LSLFormer architecture introduces an H2M module to alleviate the computational burden of self-attention via physically inspired spectral aggregation, a MCIE module to robustly extract multiscale LiDAR features, and an SLA to achieve efficient and deep cross-modal fusion. Furthermore, we propose a novel CMSC loss that ensures precise boundary delineation without compromising spectral semantics. Extensive experimental results on three public benchmark datasets demonstrated that the proposed LSLFormer achieves a superior balance between classification accuracy and computational efficiency. Our lightweight architecture consistently outperformed the compared methods in both accuracy and resource consumption. This confirms that LSLFormer successfully fuses HSI and LiDAR data by maintaining high accuracy while being inherently more efficient. The LSLFormer framework is, thus, validated as a powerful and practical solution for HSI and LiDAR fusion tasks in remote sensing and Earth observation. Future work should be directed toward exploring more efficient, generalizable, and robust fusion models for multimodal remote sensing data.

REFERENCES

[1] G. Swetha, R. Datla, C. Vishnu, and C. K. Mohan, "M2-APNet: A multimodal deep learning network to predict major air pollutants from temporal satellite images," *J. Appl. Remote Sens.*, vol. 18, no. 1, Nov. 2023, Art. no. 012005.

[2] P. Wang et al., "CIRSM-Net: A cyclic registration network for SAR and optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5610619.

[3] D. Hong et al., "Interpretable hyperspectral artificial intelligence: When nonconvex modeling meets hyperspectral remote sensing," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 2, pp. 52–87, Jun. 2021.

[4] B. Qin, S. Feng, C. Zhao, B. Xi, W. Li, and R. Tao, "FDGNet: Frequency disentanglement and data geometry for domain generalization in cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 6, pp. 10297–10310, Jun. 2025.

[5] G. Morales, J. W. Sheppard, R. D. Logan, and J. A. Shaw, "Hyperspectral dimensionality reduction based on inter-band redundancy analysis and greedy spectral selection," *Remote Sens.*, vol. 13, no. 18, p. 3649, Sep. 2021.

[6] J. Jung, E. Pasolli, S. Prasad, J. C. Tilton, and M. M. Crawford, "A framework for land cover classification using discrete return LiDAR data: Adopting pseudo-waveform and hierarchical segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 2, pp. 491–502, Feb. 2014.

[7] Y. Chen, C. Li, P. Ghamisi, X. Jia, and Y. Gu, "Deep fusion of remote sensing data for accurate classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1253–1257, Aug. 2017.

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[8] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.

[9] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[10] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[11] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[12] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2025, pp. 5998–6008.

[13] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[14] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[15] S. Wang, B. Z. Li, M. Khabsa, H. Fang, and H. Ma, "Linformer: Self-attention with linear complexity," 2020, *arXiv:2006.04768*.

[16] Z. He et al., "DADIGAN: A dual attention blocks-based disentangled iterative generative adversarial network for cloud and shadow removal on SAR and optical images," *Inf. Fusion*, vol. 125, Jan. 2026, Art. no. 103487.

[17] S. Wold, K. H. Esbensen, and P. Geladi, "Principal component analysis," *Wiley Interdiscipl. Rev., Comput. Statist.*, vol. 2, nos. 1–3, pp. 37–52, 1987.

[18] J. He, J. Li, Q. Yuan, H. Shen, and L. Zhang, "Spectral response function-guided deep optimization-driven network for spectral super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 9, pp. 4213–4227, Sep. 2022.

[19] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[20] S. Morchhale, V. P. Pauca, R. J. Plemmons, and T. C. Torgersen, "Classification of pixel-level fused hyperspectral and LiDAR data using deep convolutional neural networks," in *Proc. 8th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Aug. 2016, pp. 1–5.

[21] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 3011–3024, Jun. 2017.

[22] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[23] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[24] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[25] S. Mohla, S. Pande, B. Banerjee, and S. Chaudhuri, "FusAtNet: Dual attention based SpectroSpatial multimodal fusion network for hyperspectral and LiDAR classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 416–425.

[26] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[27] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.

[28] Y. Zhang, Y. Peng, B. Tu, and Y. Liu, "Local information interaction transformer for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1130–1143, 2023.

[29] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[30] X. Zhao et al., "Fractional Fourier image transformer for multimodal remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 2314–2326, Feb. 2024.

[31] Z. Lin, Y. Chen, X. Zhao, and G. Wang, "Spectral–spatial classification of hyperspectral image using autoencoders," in *Proc. 9th Int. Conf. Inf., Commun. Signal Process.*, Dec. 2013, pp. 1–5.

[32] D. Sinha and M. El-Sharkawy, "Thin MobileNet: An enhanced MobileNet architecture," in *Proc. IEEE 10th Annu. Ubiquitous Comput., Electron. Mobile Commun. Conf. (UEMCON)*, Oct. 2019, pp. 280–285.

[33] C. Zhang, Y. Xu, and Y. Shen, "CompConv: A compact convolution module for efficient feature learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 3006–3015.

[34] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[35] H. Cheng, M. Zhang, and J. Q. Shi, "A survey on deep neural network pruning: Taxonomy, comparison, analysis, and recommendations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10558–10578, Dec. 2024.

[36] J. Xu, Z. Li, B. Du, M. Zhang, and J. Liu, "Reluplex made more practical: Leaky ReLU," in *Proc. IEEE Symp. Comput. Commun. (ISCC)*, Jul. 2020, pp. 1–7.

[37] Y. Wu and K. He, "Group normalization," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, vol. 128, 2018, pp. 3–19.

[38] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.

[39] P. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, "MUUFL Gulfport hyperspectral and LiDAR airborne data set," Univ. Florida, Gainesville, FL, USA, Tech. Rep. REP-2013-570, 2013.

[40] B. Rasti, P. Ghamisi, and R. Gloaguen, "Hyperspectral and LiDAR fusion using extinction profiles and total variation component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3997–4007, Jul. 2017.

[41] B. Rasti et al., "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.

[42] K. Ni, Z. Li, C. Yuan, Z. Zheng, and P. Wang, "Selective spectral–spatial aggregation transformer for hyperspectral and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[43] S. Mehta and M. Rastegari, "Separable self-attention for mobile vision transformers," 2022, *arXiv:2206.02680*.

[44] S. F. C. Soares, A. A. Gomes, M. C. U. D. Araújo, A. R. G. Filho, and R. K. H. Galvão, "The successive projections algorithm," *TrAC Trends Anal. Chem.*, vol. 42, pp. 84–98, Jan. 2012.

[45] S. Ranjan, D. R. Nayak, K. S. Kumar, R. Dash, and B. Majhi, "Hyperspectral image classification: A k-means clustering based approach," in *Proc. 4th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, Jan. 2017, pp. 1–7.

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